

# Saurabh Kumar

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## SUMMARY

Computer Science student with expertise in full-stack web development and artificial intelligence. Proficient in building production-ready applications using MERN stack, TensorFlow, and cloud technologies. Active open-source contributor with demonstrated success in developing scalable solutions achieving 95% model accuracy and handling 10,000+ daily API requests.

## EDUCATION

### Dr. APJ Abdul Kalam Technical University

*Bachelor of Technology in Computer Science and Engineering; CGPA: 7.39/10.0*

Lucknow, India

*Aug. 2023 – May 2027*

### Uttar Pradesh Board (UPMSP)

*Secondary School Certificate (Class X); Percentage: 87%*

Uttar Pradesh, India

*May 2020*

## EXPERIENCE

### Open Source Contributor

*Various Open Source Projects*

Remote

*Jan. 2024 – Present*

- Contributing to React.js related ecosystem projects by fixing bugs, improving documentation, and implementing new features.
- Collaborated with maintainers of popular npm related packages to enhance functionality and resolve critical issues affecting 10,000+ users.
- Submitted pull requests to data science related repository for improving model performance and adding new examples.
- Active participant in GitHub Discussions, providing technical support and code reviews for community members.
- Maintained personal open-source projects with 20+ repositories focusing on web development and machine learning tools.

## TECHNICAL SKILLS

**Programming Languages:** JavaScript, Python, C++, HTML5, CSS3, SQL

**Frontend Technologies:** React.js, Vite, Tailwind CSS, Responsive Design, Single Page Applications (SPA)

**Backend Technologies:** Node.js, Express.js, RESTful APIs, JWT Authentication, Session Management

**Databases:** MongoDB, PostgreSQL, MySQL, MongoDB Atlas, Database Design, Query Optimization

**Machine Learning:** TensorFlow, Keras, Google Gemini AI, Deep Learning, CNN, NLP, Computer Vision

**DevOps and Tools:** Git, GitHub, Docker, Vercel, Render, AWS, Postman, Jupyter Notebook, npm, Nodemon

**Core Competencies:** Data Structures, Algorithms, System Design, OOP, Agile Methodology, Cloud Deployment

## PROJECTS

### StudyMate – AI-Powered Study Platform | *React, Node.js, MongoDB, Gemini AI*

Dec. 2024

- Engineered full-stack educational platform serving 20+ active users with AI-powered study assistance using MERN stack and Google Gemini API.
- Implemented intelligent chatbot for personalized learning with automated document summarization supporting PDF and DOCX formats.
- Developed comprehensive analytics dashboard featuring streak tracking and progress visualization using Recharts library.
- Integrated secure OTP-based email verification system using Resend API with bcrypt password encryption and session management.
- Established DDoS protection with rate limiting (3-100 requests per endpoint) and security headers using Helmet.js middleware.
- Deployed scalable architecture on Vercel (frontend) and Render (backend) with MongoDB Atlas for reliable data persistence.
- Links:** GitHub

### Brain Tumor Detection System | *React, Node.js, Python, TensorFlow, Keras*

Nov. 2024

- Developed medical AI diagnostic system for brain tumor detection from MRI scans using Convolutional Neural Networks (CNN).
- Trained deep learning model achieving 95%+ accuracy on medical imaging dataset using TensorFlow and Keras frameworks.
- Created RESTful API with Express.js enabling real-time MRI image analysis with confidence score predictions and result visualization.
- Implemented secure hospital authentication system with bcrypt password hashing and role-based session access control.
- Integrated Python ML model with Node.js backend using child processes for seamless prediction workflow and error handling.
- Designed responsive React interface with drag-and-drop MRI upload, real-time preview, and detailed analysis result display.
- **Links:** GitHub

### **HavenTo – Accommodation Booking Platform** | *React, Node.js, MongoDB, Express*

Oct. 2024

- Built full-stack booking platform handling 500+ registered users with property listings and comprehensive reservation management.
- Developed RESTful APIs processing 10,000+ daily requests with robust error handling, input validation, and response optimization.
- Implemented user authentication, favorites system, and booking functionality with secure JWT-based session management.
- Integrated Cloudinary API for optimized image storage, upload, and delivery with automatic compression and caching.
- Designed responsive user interface with React Router enabling seamless navigation and enhanced user experience across devices.
- Deployed production application on Vercel (frontend) and Render (backend) with MongoDB Atlas for scalable data storage.
- **Links:** GitHub | Live Demo

## **ACHIEVEMENTS**

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Contributed to 5+ open-source projects including React.js ecosystem and TensorFlow.js with merged pull requests. Developed and deployed 3 production-ready full-stack applications on cloud platforms (Vercel, Render, AWS). Achieved 95%+ accuracy in medical AI model for brain tumor detection from MRI scans using deep learning techniques. Built scalable RESTful APIs handling 10,000+ daily requests with optimized performance and comprehensive error handling. Solved 100+ algorithmic problems on LeetCode and GeeksforGeeks demonstrating strong problem-solving abilities. Participated in college hackathons and technical competitions, securing recognition for innovative technology solutions.

## **CERTIFICATIONS**

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Full Stack Web Development – Online Course (2024)  
Machine Learning Fundamentals – Online Course (2024)  
Data Structures and Algorithms – GeeksforGeeks (2023)